OPIHI Data Cleaning: Handoff Guide

Nov 3, 2026 · @Cam

Status at a glance

The OPIHI data are cleaned and verified: 20 raw workbooks became one master dataset of **22,588 detections** across **252 surveys** and **7,216 samples**, and all 13 verification checks passed against an independent calculation.

Item	Count
Raw Excel workbooks read	20 (59 tabs; 55 also saved as CSV)
Surveys (site visits) in the master	252
Samples (quadrats + student groups)	7,216
Detections (rows in `opihi_master.csv`)	22,588
Sites	43 sites in the master (44 in the lookup) on 6 islands
Taxa	215 clean names, 137 analysis taxa
Records excluded (all logged)	427
Trips in compiled files with species data not in the raw files	0

Done: import of all raw formats, cleaning, deduplication, site and taxon standardisation, anonymous teacher IDs, exclusions log, QC flags, data dictionary, cross-check against compiled files, AI Developer Log.

Remaining: choose the focal taxon, the reusable function with tests, two figures, a formatted table, README and `.gitignore`, then render and publish (details in the last section).

Project folder layout and how to rerun

Everything runs from one Quarto report, `Scripts/opihi_report.qmd`, which reads the raw Excel files and rewrites every processed file from scratch; nothing is edited by hand.

```
AI data/                          project folder (any name, any location)
├── OPIHI.Rproj                   open this first; it marks the project root
├── Data/
|   ├── Raw/                      the 20 original Excel files, never modified
|   ├── Lookups/                  site_crosswalk.csv, taxon_crosswalk.csv
(editable inputs)
|   └── Processed/                every output the code writes
|       ├── raw_csv/              each raw tab saved as CSV
|       └── private/              teacher_key_PRIVATE.csv (real names, never
share)
├── Scripts/
|   ├── opihi_report.qmd          the report: runs the whole pipeline
|   ├── helpers.R                 shared reading/date/time utilities
|   ├── import_legacy.R           reads 2004-2020 files (Family A)
|   ├── import_modern.R           reads 2024-2026 files (Family B)
|   ├── import_classroom.R        reads classroom Field Study files (Family C)
|   └── clean_build.R             cleaning, deduplication, master tables
└── Output/                       figures and tables (to come)
```

To rerun:

1. Double-click `OPIHI.Rproj` to open the project in RStudio.
2. Open `Scripts/opihi_report.qmd`.
3. Run the setup chunk. If packages are missing, it stops and prints the exact `install.packages(...)` line to run.
4. Run all chunks top to bottom, or press **Render** to build the HTML report.

Paths use the `here` package, so the code works on any computer without editing paths. Close any output CSV that is open in Excel before rerunning; Windows locks open files and R cannot overwrite them.

The raw data: four format families

The 20 workbooks use four different layouts, so the code has one importer per layout ("family"), not one per file. Families A, B and C are primary data; Family D files are copies and summaries used only to cross-check.

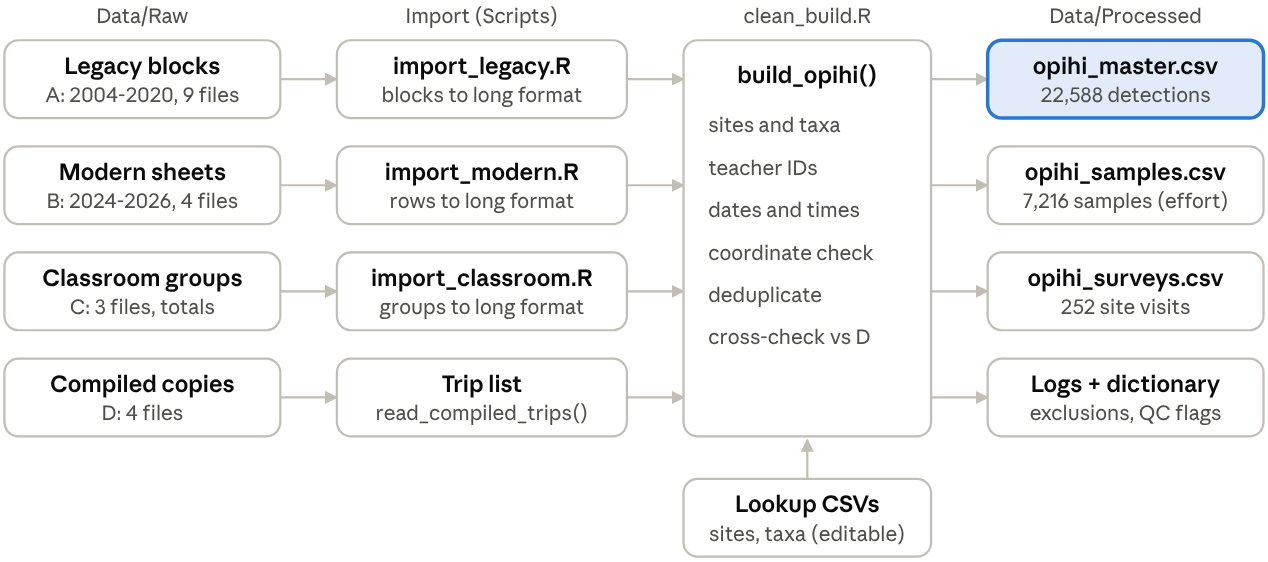
Family	Files	Layout	Resolution	In the master?
A. Legacy blocks (2004-2020)	`OPIHI data 2004`, `2005`, `2006 and 2007`, `2008 might be mis-type...`, `2016`, `2017`, `2018`, `2019`, `2020`	Each field trip is a block of columns side by side: metadata rows on top, one row per taxon, one column per quadrat	Quadrat	Yes: 227 surveys
B. Modern row-per-quadrat (2024-2026)	`OPIHI 2024 datasheet`, `OPIHI BIOL 499 2025 LT data`, `OPIHI 2026 Longitudinal Monitoring Data`, `RAW Mauliola OPIHI Data 2026`	One row per quadrat, one column per taxon (`Quadrat PtCounts` tab) plus a `Metadata` tab	Quadrat	Yes: 21 surveys
C. Classroom Field Study	`Copy of OPIHI Field Study 2023...`, `Copy of OPIHI Field Study 2025...`, `OPIHI Field Study 2026 - Waipuilani Data`	One column per student group (P1G1, B4G2...), one row per taxon; values are group totals	Student group	Yes, as `group_total`: 4 surveys
D. Derived / compiled	`OPIHI Data - Reformated`, `Playing around... Jo March 2022`, `Sandy Beach Only_`, `Copy of OPIHI 2024 datasheet`	Trip totals, subsets and recalculations of A and B	Trip	No: cross-check only

Each value in A, B and C is a **point-intercept count**: the number of the 25 points in a quadrat that landed on that taxon or substrate. A quadrat therefore holds up to 25 points; a student group of 7 quadrats holds up to 175.

What the code does

The pipeline runs in five stages: read every raw tab exactly as laid out, import each family into the same long format, standardise names with the lookup tables, remove duplicates and apply QC rules, then write the processed files.

Every raw format flows through one build step into the processed files



OPIHI pipeline · 4 raw formats, 1 build step, processed files

Read left to right: each raw format has its own importer, everything meets in `build_opihi()` with the two lookup tables, and the three analysis files come out the right side.

Script	Function	What it does
`helpers.R`	`read_sheet_matrix()`	Reads one Excel tab as plain text, anchored at cell A1 so row and column numbers match Excel. Nothing is type-guessed or altered.
	`excel_to_date()`, `excel_to_time()`	Convert Excel date serial numbers and typed dates to real dates; times to 24-hour `HH:MM`.

Script	Function	What it does
	`export_raw_csv()`	Saves every readable tab to `Data/Processed/raw_csv/`; logs tabs it cannot read (4 chart-only tabs).
	`list_raw_xlsx()`	Lists raw Excel files, ignoring Excel's temporary `~$` lock files.
`import_legacy.R`	`parse_legacy_file()`, `import_legacy()`	Finds each field-trip block by its `Organisms` header, reads its metadata by label name, turns the taxon × quadrat grid into one row per non-empty cell.
	`check_legacy_totals()`	Re-sums the parsed counts and compares them with each block's own `TOTALS:` section (3,521 totals, 0 mismatches).
`import_modern.R`	`parse_modern_file()`, `import_modern()`	Reads the `Quadrat PtCounts` tab (one row per quadrat) and the `Metadata` tab; applies the row rules (template, example, summary rows) and checks each row against its `Total Points`.
`import_classroom.R`	`parse_classroom_tab()`, `import_classroom()`	Finds the student-group columns, the date above each block of groups, and each group's `TOTAL Points` (÷ 25 = its number of quadrats).
`clean_build.R`	`key_site()`, `key_taxon()`	Reduce a raw name to a matching key: lower case, single spaces; for sites also ʻokina, apostrophes and kahakō removed.
	`standardize_sites()`, `standardize_taxa()`	Look each key up in the crosswalk CSVs; unmatched names are listed with the closest existing name as a suggestion, never matched automatically.

Script	Function	What it does
	`split_names()`, `tidy_org()`	Tidy people and school names before anonymous teacher IDs are assigned.
	`fix_field_time()`	Applies the time rule: hours 01-05 are assumed to be afternoon (+12 h) and flagged.
	`haversine_km()`	Distance between coordinates, used to check that each trip's coordinates match its site label.
	`read_compiled_trips()`	Reads the trip list from the compiled (Family D) files for the cross-check.
	`build_opihi()`	Combines the three imports and does all cleaning, deduplication, ID assignment and cross-checking; returns every output table.

`opihi_report.qmd` calls these in order, prints a verification table after each step, and writes the outputs. Each function has a comment header in its script that explains its inputs and outputs.

Cleaning rules applied

Every change is made by a written rule in the code, and every removed or changed record is listed in `opihi_exclusions_log.csv` or `opihi_qc_flags.csv`.

Topic	Rule	Effect
Raw data	Raw files are only read, never edited	Original data preserved
Blank cells	A blank cell in a filled quadrat = 0 (not detected); a quadrat with every taxon blank = not entered	109 unentered rows excluded (`no_taxon_data`)

Topic	Rule	Effect
Template rows	Rows with no date and all taxa blank or 0	309 excluded (`blank_template_row`)
Example rows	Any row containing "EXAMPLE" (student templates)	1 excluded (`example_row`)
Summary rows	"Column Totals" rows	1 excluded (`summary_row`)
Legacy block with no date, site or school	Cannot be placed in time or space	1 excluded (`no_survey_metadata`)
Taxon name that is only a number	Data-entry error	6 cells excluded (`invalid_taxon_name`)
Dates	Data-row date wins over a sheet's metadata date; an unreadable date between two rows with the same date and site takes that date	Flagged `date_inferred` , `metadata_date_mismatch`
Times	24-hour `HH:MM` ; legacy placeholder 00:00-01:00 removed; hours 01-05 assumed PM	Flagged `placeholder_time_removed` , `time_assumed_pm`
Spreadsheet errors	`#REF!` and similar become blank (NA)	
Site names	One name per site via `site_crosswalk.csv` : Hawaiian name first, no ʻokina or kahakō, English name kept as alias (e.g. Diamond Head → Leahi, Sand Island → Mauliola, Barbers Point → Kalaeloa)	66 spellings → 44 sites
Site vs coordinates	Coordinates > 5 km from the labelled site but matching another site's coordinates (< 0.2 km) → relabelled	3 trips corrected (`site_corrected_by_coordinates`)

Topic	Rule	Effect
Taxon names	Spelling, format and outdated names fixed via `taxon_crosswalk.csv`	300 spellings → 215 clean taxa in the master
Sheet totals	Where a typed `Total Points` disagrees with the row's cells, the cells are kept	Flagged `total_points_mismatch`
Point sums	A quadrat summing to more than 25 points is kept for review	Flagged `points_over_25`
Duplicates	Identical survey content in two places kept once; identical quadrats within a survey kept once; same site and date in two files flagged	0 exact duplicates found
Derived tabs	PercentCover, Grouped sums and the compiled files are calculated from the raw data	Not imported; used only to cross-check
Transect data	Point-intercept transect tabs	Not in the master (quadrat data only)
Teachers	Legacy School and modern Team Leader replaced by anonymous `T###` IDs; several on one survey joined with `;`	Name key kept only in `private/`
Students	"Data Entry By" and student names	Not in any output

Output files and columns

Three files hold the data for analysis: **`opihi_master.csv`**, **`opihi_samples.csv`** and **`opihi_surveys.csv`**. They join on `survey_id` and `sample_id`. The rest document what was done. All are in `Data/Processed/`.

File	One row per	Use it for	Priority
`opihi_master.csv`	detection: a taxon recorded in one sample (count > 0)	Where and when each taxon was recorded; cover	**Main analysis file**
`opihi_samples.csv`	quadrat or student group, including empty ones	Sampling effort: the denominator for detection frequency	**Essential with the master**
`opihi_surveys.csv`	survey (one site visit from one source)	Site, date, coordinates and effort per visit; maps	**Essential for maps and effort**
`opihi_data_dictionary.csv`	column	What each column means	Reference
`opihi_exclusions_log.csv`	excluded record	What was removed and why	Documentation
`opihi_qc_flags.csv`	flagged record	What was kept but needs care	Documentation
`opihi_compiled_crosscheck.csv`	trip in a compiled file	Proof nothing was lost from the compiled files	Documentation
`opihi_unmatched_names.csv`	name missing from a lookup	Names to add to the crosswalks (currently empty)	Maintenance
`raw_csv/*.csv`	raw tab	The raw data in open CSV format	Archive

File	One row per	Use it for	Priority
`private/teacher_key_PRIVATE.csv`	teacher	Real name → anonymous ID	**Private: never share or commit**

`opihi_master.csv` (22,588 rows)

Column	Meaning
`survey_id`	Survey ID (`S0001` ...), numbered by date; joins to `opihi_surveys`
`sample_id`	Sample ID (`S0001-001` ...); joins to `opihi_samples`
`date` , `year`	Survey date (YYYY-MM-DD) and year; undated classroom files have a year only
`site_name`	Standard site name, Hawaiian first, no ʻokina or kahakō
`site_alias_english`	English name where one is common (e.g. Diamond Head for Leahi)
`island`	Oahu, Kauai, Molokai, Lanai, Maui or Hawaii
`latitude` , `longitude`	Decimal degrees; recorded on 2004-2020 datasheets only
`teacher_id`	Anonymous teacher, school or team-leader ID(s), e.g. `T012` or `T003;T007`
`resolution`	`quadrat` (one row per quadrat) or `group_total` (classroom group totals)
`sample_type`	`quadrat` or `student_group`
`group_label`	Student group label (classroom files only), e.g. `P1G3`
`transect` , `quadrat` , `point_m`	Transect number, quadrat number (legacy only), position along the transect in metres
`n_quadrats`	1 for a quadrat; number of quadrats in a student group

Column	Meaning
`taxon_raw`	Name exactly as written on the datasheet
`taxon_clean`	Standard name: spelling fixed, current taxonomy
`analysis_taxon`	Coarser name (usually genus) for consistent comparison across years
`category`	Algae, Cyanobacteria, Invertebrate, Coral, Substrate or Unknown
`count`	Points hit by this taxon in the sample (always > 0)
`points_possible`	25 × `n_quadrats`; `count / points_possible` = proportion cover
`sample_flag`	QC flags on the sample, `;`-separated
`source_family`, `source_cell`	Raw format family (A, B, C) and the exact raw file, row and column the value came from

`opihi_samples.csv` (7,216 rows)

Column	Meaning
`sample_id`, `survey_id`	IDs, as in the master
`sample_type`, `group_label`, `transect`, `quadrat`, `point_m`, `n_quadrats`	As in the master
`points_possible`	25 × `n_quadrats`
`point_sum`	All counts in the sample added up (25 expected per quadrat in modern files; often less in legacy files, where bare substrate was not always recorded)
`n_taxa`	Number of different non-substrate taxa in the sample
`sample_flag`	QC flags
`sample_key`	Internal link back to the raw file

`opihi_surveys.csv` (252 rows)

Column	Meaning
`survey_id`, `date`, `year`, `site_name`, `site_alias_english`, `island`, `latitude`, `longitude`, `teacher_id`, `resolution`	As in the master
`source_family`, `source_file`	Which raw file the survey came from
`start_time`, `stop_time`, `low_tide_time`	24-hour `HH:MM`
`n_transects_reported`, `n_quad_per_t_reported`, `transect_length_m`, `point_spacing_m`	Effort written on the legacy datasheet (may differ from what was actually recorded)
`n_samples`, `n_quadrats`	Samples and quadrats actually in the cleaned data: the survey's sampling effort
`src_key`	Internal link back to the raw file or block

Documentation files

File	Columns
`opihi_exclusions_log.csv`	`level` (survey, row, sample, observation), `source_file`, `source_ref` (exact raw location), `reason` (rule name), `detail`
`opihi_qc_flags.csv`	`level`, `ref` (survey key or sample ID), `flag` (rule name), `detail`
`opihi_compiled_crosscheck.csv`	`source_file`, `excel_row`, `date`, `site_raw`, `site_name`, `has_species_data`, `status` (234 trips in master; 18 compiled-only trips with no species data)
`opihi_unmatched_names.csv`	`type` (site or taxon), `raw`, `suggestion`
`opihi_data_dictionary.csv`	`table`, `column`, `type`, `description`

The lookup tables in `Data/Lookups/` are inputs, not outputs. `site_crosswalk.csv` has `site_key`, `site_name`, `site_alias_english`, `island`, `review_note`, `example_raw`; `taxon_crosswalk.csv` has `taxon_key`, `taxon_clean`, `analysis_taxon`, `category`, `review_note`, `example_raw`. To fix or add a name, edit these files and rerun.

Pulling out locations for a map

Yes: every one of the 43 sites in the master has coordinates, so all of them can be mapped. Use **`opihi_surveys.csv`** for a map of where sampling happened, and join **`opihi_master.csv`** to it for a map of where a taxon was found.

Three things to know first:

- **Coordinates come only from the 2004-2020 datasheets.** Modern (2024-2026) and classroom surveys have blank `latitude` / `longitude`, but they are at sites that also have legacy visits, so each site's coordinates can be taken from those.
- **Use one coordinate per site, not per visit.** Coordinates were typed per trip and vary slightly; the median per site gives one stable point.
- **Two classroom surveys (2023, 2025) have no site.** Drop rows with a blank `site_name` before mapping.

A site table for mapping, from the processed files:

```
library(dplyr); library(readr)
surveys <- read_csv(here::here("Data", "Processed", "opihi_surveys.csv"))

sites <- surveys |>
  filter(!is.na(site_name)) |>
  group_by(site_name, site_alias_english, island) |>
  summarise(latitude   = median(latitude, na.rm = TRUE),
            longitude  = median(longitude, na.rm = TRUE),
            n_surveys  = n(),
            n_quadrats = sum(n_quadrats),
            first_year = min(year, na.rm = TRUE),
            last_year  = max(year, na.rm = TRUE),
            .groups = "drop")
```

To map where one taxon was recorded, count the surveys where it was detected and join that to `sites`:

```
master <- read_csv(here::here("Data", "Processed", "opihi_master.csv"))

taxon_sites <- master |>
  filter(analysis_taxon == "Acanthophora spicifera", !is.na(site_name)) |>
  group_by(site_name) |>
  summarise(n_surveys_detected = n_distinct(survey_id), .groups = "drop") |>
  right_join(sites, by = "site_name") |>
  mutate(n_surveys_detected = coalesce(n_surveys_detected, 0L))
```

`right_join` keeps sites where the taxon was never recorded, so they appear on the map as sampled-but-absent, which is part of the answer. These are suggestions to adapt; a site table can also be added to the pipeline as its own output file (`opihi_sites.csv`) when the figures are built.

Detection frequency: accounting for effort

Detection frequency for a taxon at a survey = **quadrats where it was recorded ÷ quadrats sampled**. The numerator comes from `opihi_master.csv`; the denominator must come from `opihi_samples.csv`, because the master lists only detections and quadrats without the taxon never appear in it.

```
samples <- read_csv(here::here("Data", "Processed", "opihi_samples.csv"))

detected <- master |>
  filter(resolution == "quadrat", analysis_taxon == "Turbinaria ornata") |>
  distinct(sample_id) |>
  mutate(detected = TRUE)

det_freq <- samples |>
  filter(sample_type == "quadrat") |>
  left_join(detected, by = "sample_id") |>
  mutate(detected = coalesce(detected, FALSE)) |>
  group_by(survey_id) |>
  summarise(n_quadrats = n(), n_detected = sum(detected),
            detection_freq = n_detected / n_quadrats, .groups = "drop") |>
  left_join(select(surveys, survey_id, date, year, site_name, island), by =
"survey_id")
```

Rules of thumb:

- **Quadrat data only** (`resolution == "quadrat"`). Classroom group totals cannot say how many quadrats a taxon was in.
- **Use `analysis_taxon` to compare across years.** Identification detail changed over time (genus in one year, species in another); `taxon_clean` is for questions about a specific species.
- **Show effort next to the result.** A detection frequency from 5 quadrats is much less certain than one from 40. OPIHI asks teachers for 25 quadrats per trip and has used 20 as a cut-off in past analyses; a minimum like that is a reasonable filter.
- **Be careful comparing cover between eras.** Modern sheets record substrate (bare rock, sand), so quadrats sum to 25 points; legacy sheets often did not, so their sums are lower. Detection (present or absent) is comparable across eras; cover (`count / points_possible`) is less so.

Known caveats

These are not errors in the pipeline but limits of the data, or judgement calls someone should know about before drawing conclusions.

- **Judgement calls in the lookups.** Lines marked `REVIEW` in `taxon_crosswalk.csv` are calls worth a second look: *Avrainvillea amadelpha* merged into *A. lacerata*; "Cerramium"/"cerracia" read as *Ceramium*; "Gelid spp" kept only to order level; *Lyngbya majuscula* (possibly *Moorena producens* today); "Tube snail" treated as vermetids. In `site_crosswalk.csv` , "K Bay" was matched to Kahana Bay because its coordinates are identical.
- **Two classroom files have no site or date** (2023 and 2025). They are kept with their year only and cannot be mapped.
- **Some surveys were never fully entered.** Mauliola on 2026-01-25 has 26 entered quadrats and 107 blank ones; 18 trips listed in the compiled files have no species data at all.
- **Point sums differ by era.** Legacy quadrats often sum to less than 25 because bare substrate was not always recorded; quadrats over 25 (174 in legacy files, more in modern files) are kept and flagged, not removed.
- **Teacher IDs follow the names as written.** Legacy School names were not merged when spelled differently (e.g. "UH Manoa" and "University of Hawaii at Manoa" get different IDs), and early trips list "OPIHI Coordinator" as the school. Questions about teacher effects need that harmonised first.

- **Rare taxa are thin.** ʻOpihi (*Cellana*) appear in only about 12 detections, too few for a time trend on their own. Students are also known to misidentify cryptic and rare species more often.
- **Privacy.** `Data/Raw/` holds the unredacted datasheets and `Data/Processed/private/` holds the real-name key. Neither may go on public GitHub or into an external AI tool. All other outputs contain only anonymous IDs.

Remaining tasks

Five build steps remain, in this order, because steps 3 to 5 depend on the taxon chosen in step 2. The AI Developer Log is done; rendering, publishing and individual reflections come last.

Step	Task	Decision needed	Uses
2	Choose the focal taxon or group and write the question and metric precisely	Which taxon or group (e.g. invasive algae, *Turbinaria ornata*, urchins, ʻopihi) and at which level (`taxon_clean` or `analysis_taxon`)	`opihi_master.csv`
3	One reusable function with documented inputs and outputs, and a test	Detection-frequency calculator, new-datasheet checker (flags unknown names, bad point sums, duplicates in incoming data), or both	master, samples, lookups
4	Two labelled ggplot figures with captions	Which two figures, e.g. a map of sites where the taxon was found, and detection frequency over time at repeatedly sampled sites	master, samples, surveys
5	A formatted ("beautiful") table of a data subset, using the `gt` package	Which subset, e.g. a detection summary for the taxon by site	master, surveys
6	README and `.gitignore`	Repo name and team member names for the README	—

Then:

- [ ] Render `opihi_report.qmd` to HTML and confirm it builds start to finish
- [ ] Push to the course GitHub organisation, with `Data/Raw/` and `Data/Processed/private/` excluded
- [ ] Publish the HTML report
- [ ] Each team member writes an individual reflection